NTP Security

Topology

Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
line con 0
logging synchronous
exit
no ip domain-lookup
int gig 0/1
ip address 192.0.2.100 255.255.255.0
no shut
router ospf 1
```

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```
network 0.0.0.0 0.0.0.0 area 0
end
copy run star
```

INTERNET:

```
enable
conf t
host INTERNET
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 192.0.2.1 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
int lo0
ip address 1.1.1.1 255.255.255.255
ip http server
end
copy run star
```

NTP basic setup for INTERNET:

```
clock set 12:00:00 June 28 2024
conf t
clock timezone UTC 0
ntp master 3
end
```

(##Note, we need to set up NTP on the INTERNET router first before we set up NTP security.)

NTP basic setup for R1:

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```
conf t
ntp server 1.1.1.1
clock timezone EST -5
clock summer-time EDT recurring
end
```

(##Note, we need to set up NTP on R1 first before we set up NTP security.)
(##Don't forget, we will have to wait approximately 15 minutes for R1 to fully synchronize with the INTERNET router.)

Lab Tasks

1. On our INTERNET router, let's first configure our basic setup for NTP that we find above in our initial configuration commands section.
2. On R1, let's configure our basic setup for NTP that we find above in our initial configuration commands section.
3. On our INTERNET router, let's set up an authentication key with a key number of "1" and a key of "$3creT". Let's also say that this is the key that we trust on this router.
4. On the INTERNET router, let's require NTP authentication. Set this router as an NTP master with a stratum value of 3.
5. On the NTP client, which is R1, let's set up a key number of "1" and match our key to the one we set on our INTERNET router. Set this key as our trusted key for R1.
6. On R1, let's require NTP authentication. Let's say that when we are talking to the server with an IP address of 1.1.1.1, we are going to be using key number "1".
7. On our INTERNET router, let's create a standard named ACL, and let's name it "NTP-CLIENT". Also permit the outgoing IP address of R1.
8. Let's apply this ACL to our NTP configuration and set it to only serve time to R1, or our NTP-CLIENT in this case.
9. On R1, let's create a standard named ACL, and let's name it "NTP-SERVER". Finally, permit 1.1.1.1 and then apply this ACL to our NTP configuration.
10. Let's verify that we have correct time on R1 by using the "show ntp status" command.

Solution

Step 1: On our INTERNET router, let's first configure our basic setup for NTP that we find above in our initial configuration commands section.

```
INTERNET>en
INTERNET#clock set 12:00:00 June 28 2024
INTERNET#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
```

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```
INTERNET(config)#clock timezone UTC 0
INTERNET(config)#ntp master 3
INTERNET(config)#end
```

Step 2: On R1, let's configure our basic setup for NTP that we find above in our initial configuration commands section.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ntp server 1.1.1.1
R1(config)#clock timezone EST -5
R1(config)#clock summer-time EDT recurring
R1(config)#end
```

Step 3: On our INTERNET router, let's set up an authentication key with a key number of "1" and a key of "$3creT". Let's also say that this is the key that we trust on this router.

```
INTERNET#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
INTERNET(config)#ntp authentication-key ?
  <1-4294967295>  Key number

INTERNET(config)#ntp authentication-key 1 ?
  md5  MD5 authentication

INTERNET(config)#ntp authentication-key 1 md5 $3creT
INTERNET(config)#ntp trusted-key 1
```

(##Notice that we can set up more than one key number here.)

Step 4: On the INTERNET router, let's require NTP authentication. Set this router as an NTP master with a stratum value of 3.

```
INTERNET(config)#ntp authenticate
INTERNET(config)#ntp master 3
INTERNET(config)#end
```

(##This shows that we are going to be serving time from our local clock, which we have already set above.)

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Step 5: On the NTP client, which is R1, let's set up a key number of "1" and match our key to the one we set on our INTERNET router. Set this key as our trusted key for R1.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ntp authentication-key 1 md5 $3creT
R1(config)#ntp trusted-key 1
```

(##Our key has to match what we set on our INTERNET router, but our key number doesn't have to match.)

Step 6: On R1, let's require NTP authentication. Let's say that when we are talking to the server with an IP address of 1.1.1.1, we are going to be using key number "1".

```
R1(config)#ntp authenticate
R1(config)#ntp server 1.1.1.1 key 1
R1(config)#end
```

Step 7: On our INTERNET router, let's create a standard named ACL, and let's name it "NTP-CLIENT". Also permit the outgoing IP address of R1.

```
INTERNET#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
INTERNET(config)#ip access-list standard NTP-CLIENT
INTERNET(config-std-nacl)#permit 192.0.2.100
```

Step 8: Let's apply this ACL to our NTP configuration and set it to only serve time to R1, or our NTP-CLIENT in this case.

```
INTERNET(config-std-nacl)#ntp access-group serve-only NTP-CLIENT
INTERNET(config)#end
```

Step 9: On R1, let's create a standard named ACL, and let's name it "NTP-SERVER". Finally, permit 1.1.1.1 and then apply this ACL to our NTP configuration.

```
R1#conf t
Enter configuration commands, one per line.  End with
CNTL/Z.
R1(config)#ip access-list standard NTP-SERVER
R1(config-std-nacl)#permit 1.1.1.1
R1(config-std-nacl)#exit
R1(config)#ntp access-group peer NTP-SERVER
```

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```
R1(config)#end
```

Step 10: Let's verify that we have correct time on R1 by using the "show ntp status" command.

```
R1#show ntp status
Clock is synchronized, stratum 4, reference is 1.1.1.1
nominal freq is 1000.0003 Hz, actual freq is 999.6496 Hz, precision is 2**13
ntp uptime is 714200 (1/100 of seconds), resolution is 1001
reference time is EA293E7D.A698ED72 (09:51:57.650 EDT Fri Jun 28 2024)
clock offset is -6.6187 msec, root delay is 9.81 msec
root dispersion is 7948.40 msec, peer dispersion is 4.41 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000350626 s/s
system poll interval is 64, last update was 467 sec ago.
```

(##R1 has a stratum of 4 because the master has a stratum of 3.)
(##We can see that 1.1.1.1 is our reference clock.)

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